

Mohit Manna

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EDUCATION

Purdue University

Dec 2027

BS Computer Engineering

Minors: Math & Finance

- **Coursework:** Operating Systems · Distributed Systems · Data Structures & Algorithms · Computer Architecture
- **Orgs/Awards:** USACO Silver · ML @ Purdue · Data Mine · Boiler Blockchain · Embedded Systems

EXPERIENCE

IBM | *Software Engineer Intern*

May – Aug 2026

- Launched a FastAPI LLM triage service for 42 engineers, running 10K+ analyses and cutting diagnosis time by 80%
- Rate-limited Azure OpenAI calls with Redis across 4 Kubernetes pods, sustaining 35 concurrent analyses within quota
- Ingested webhooks across 32 CI workflows on asyncio, writing Pydantic-validated results to PostgreSQL at p95 under 5s
- Indexed 4,500 past CI failures in pgvector with an HNSW cosine index, pulling the 5 nearest matches in under 5 ms

Handshake | *AI Engineer*

Jan – Apr 2026

- Built an RLHF pipeline for code LLMs, training on 50K+ preferences and improving pass@1 by 18%
- Engineered a sandboxed code-eval harness, scoring 250K+ generations at 7.4x higher throughput
- Optimized LLM serving via vLLM and AWQ, cutting p95 latency by 42% on 8x A100 GPUs

Caterpillar | *Machine Learning Intern*

Aug – Dec 2025

- Boosted supply-chain forecast accuracy by 12% with a PyTorch Temporal Fusion Transformer tuned over 60 Optuna trials
- Scaled TorchScript batch inference with Docker and Kubernetes, cutting runtime to 2.85 min across 520 SKUs
- Automated a retraining pipeline on drift alerts, cutting over from a 4-week shadow deploy to 20 Angular dashboards

Stealth AI Startup | *Software Engineer Intern*

Jun – Aug 2025

- Shipped React dashboard with GraphQL and WebSockets, handling 500+ concurrent connections across 10K data points
- Trained a scikit-learn AI model on 50K+ market records using pandas, reaching 89% backtesting accuracy
- Secured 40+ Express endpoints with JWT authentication and PostgreSQL RLS, validated by 50+ Jest integration tests

PROJECTS

PulseKV — *Concurrent Key-Value Storage Engine in C*

- Sharded mutex-guarded hash buckets on 16-thread epoll server, sustaining 25K+ req/sec across 500 clients at <5 ms p99
- Designed checksummed append-only persistence with batched log replay, cutting restart time 60% across 1M+ keys

Fintrak – *AI-Powered Finance Tracker*

- Streamed LLM completions over 90-day histories, powering a financial copilot tracking \$180K+ in budgets for 200+ users
- Architected AI cron pipelines detecting spending spikes above 2× 30-day baselines, delivering 600+ daily budget nudges

Research Assistant – *Photonic Simulation Lab*

- Accelerated 10,000+ waveguide simulations via Python multiprocessing and HDF5 caching, cutting manual effort by 18%
- Placed 1st among 200 participants at the Purdue Research Symposium for 99.7%-accurate waveguide simulations

SKILLS

- **Languages:** Python · TypeScript · C · C++ · Java · SQL · Go
- **Frameworks:** React · Node.js · Express · Pydantic · FastAPI · Flask · Angular · PyTorch · GraphQL
- **Tools:** AWS · Linux · Git · Docker · Kubernetes · Azure · PostgreSQL · Redis · GitHub Actions · OpenTelemetry